SectraTEK

AI FOR CYBERSECURITY AND DIGITAL FORENSICS · LABS 07 AND 08

Talk to the Evidence



Turn the forensic agent into a chatbot: ask a device image (Lab 07) or an Android bug report (Lab 08) in plain English, and get answers that cite their source.

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THE IDEA

From a lab you run, to a chatbot you talk to

In the lab you built the agent step by step. Now we wrap that same agent in a chat window, so you (or a student, or a colleague) can just ask questions and watch it work.

IN THE LAB

You call run_agent(...)

Run cells, read the trace, build the tools by hand. Great for learning how it works.



AS A CHATBOT

You just type a question

A chat window drives the same agent. Good for demos, quick checks, and letting others try it.

TWO CHATBOTS

Two things you can point it at

Lab 07: device image

Ask the Device

Walks a full Android extraction and answers with the exact file it read.

"Which accounts are on this phone?" → reads the account database, cites accounts.db.

Lab 08: bug report

Ask the Bug Report

Reads an Android bug report, which you can pull from your own phone, and cites the section.

"Which apps are installed?" → reads the package section, cites it.

LAB 08: WHY A BUG REPORT

Light enough to use your own phone

A full extraction needs special tools. An Android bug report is just a text file, so anyone can make one.

Lightweight

One sectioned text file: device info, apps, battery, Wi-Fi, accounts, logs.

From your own phone

Enable Developer Options, then run adb bugreport. No forensic kit needed.

Easy to search

The agent lists sections, reads one, or searches the whole report.

```
adb bugreport  mybugreport      # produces mybugreport.zip, then upload it to the chatbot
```

HOW IT WORKS

Same agent, with a chat window on top

The tools do the work

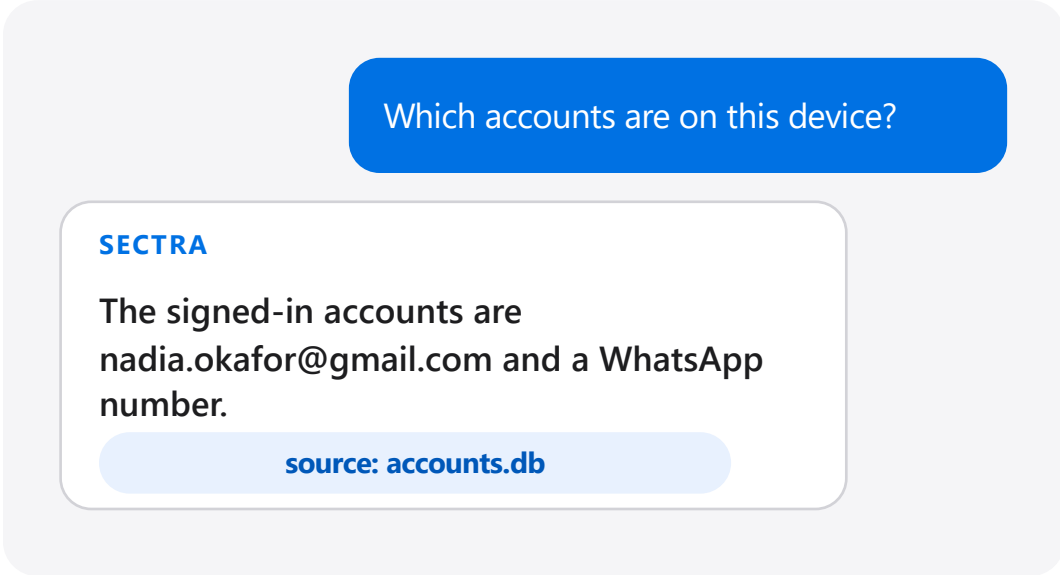
The agent lists, reads, and searches the evidence with read-only tools.

The loop runs the same

Think, act, observe, repeat, as in the lab, from the packaged toolbox agentkit.py.

Every answer is cited

It names the file or section it used, so you can open and check it.



THE EVIDENCE CHECK

Every answer is checked before you see it

Giving an AI access to the evidence limits what it can look at, not what it can say. So the toolbox code, not the AI, checks each answer against what the tools really returned.



Did it look?

At least one tool must have run for the question. A question asked back to you is allowed.



Is every detail real?

Every file name, IP address, process ID, and memory address (0x...) must appear in what a tool returned.



Is the citation honest?

The answer's cited source cannot name a tool that never ran.

IF IT FAILS

It goes back to the AI to check with a tool or drop the claim (an evidence_check step in the trace). After two tries it is shown marked UNVERIFIED: a lead, not a finding. It catches invented details, not a wrong reading of real evidence.

TWO WAYS TO RUN EACH

A web window, or right in the notebook

UI version

```
chatbot_ui.ipynb
```

Launches a real chat window (built with Gradio). In Colab you get a link you can open and share.

Best for demos and for people who just want to ask.

Jupyter version

```
chatbot_jupyter.ipynb
```

Call ask("...") in a cell and read the reasoning and the cited answer, no web UI.

Best for working through questions one at a time.

RUN IT ANYWHERE

Hosted in the cloud, or private on your PC

hosted model

In Colab

Open the notebook, paste your free Groq key, and go. Nothing to install.

- No setup
- Free model
- Shareable link

local Qwen (Ollama)

On your PC

Run a Qwen model locally with Ollama. No API key, and nothing leaves your machine.

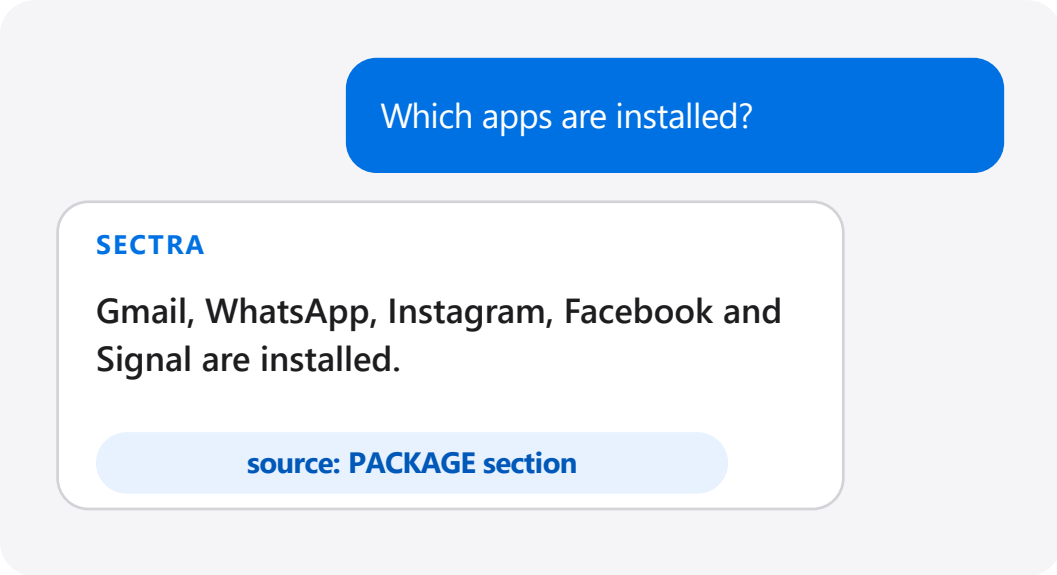
- No key
- Fully offline
- Private

USE IT

Ask in plain English, get a cited answer

THINGS TO ASK

- Which email accounts are signed in on this device?
- Which apps are installed?
- What is the most recent text that was sent?
- What Wi-Fi networks are saved?
- Any errors or crashes in the log?



Every answer names the file or section behind it, so you can verify it yourself.

YOUR TURN

Open a chatbot and ask it something

1	**Open a chatbot notebook**	Device: chatbot_ui.ipynb or chatbot_jupyter.ipynb. Bug report (Lab 08): bugreport_chatbot_ui.ipynb.
2	**Install and save the toolbox**	Steps 1 and 2: install the helpers, then save agentkit.py (tools, loop, evidence check).
3	**Connect to the model**	Paste your free Groq key (the same as Labs 01-06), or set PROVIDER to ollama for no key.
4	**Launch, or ask**	UI: Step 5 prints a link to the chat window. Jupyter: Step 4 calls ask(...), Step 5 is yours.
5	**Ask and check**	Open the cited source for every answer. Treat UNVERIFIED as a lead, not a finding.

Every step explains its Purpose, Input, How it works, and Output. The bug-report chat uses a sample, or your own upload.

SectraTEK

Ask, and check the source.

The chatbot makes the agent conversational, and still shows the exact file or section behind every answer.

